

Curriculum Vitæ

Kevin Kramer

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Professional Summary: Scientific Software Engineer and Researcher with a PhD in Physics and a track record of optimizing large-scale parallel computations on HPC clusters. Experienced in performance tuning and parallelization (MPI, SLURM-managed supercomputers) of physical simulations and calibration algorithms, alongside developing reproducible, containerized data workflows and scientific software packages (Python, C++, R). Proven track record in open-source development (growR, data-slicer) and bridging the gap between complex scientific research and robust, high-performance software engineering.

Work and Education

2023 – 2025 Scientist (Climate Modeling & Software Engineering)

Modeling of climate change impacts on Swiss agriculture

Agroscope: Federal Research Institute for Agronomy

Software Engineering: Developed and maintained the [growR R package](#), providing a modular, well-documented framework for vegetation modeling. **Data**

Workflow Development: Designed automated (Bash/Python/R) and containerized data pipelines for carrying out reproducible and efficient climate impact simulations. **High Performance Computing:** Optimized and parallelized a Bayesian calibration approach and deployed it on a SLURM-managed HPC environment, reducing computation time by two orders of magnitude.

Visualization and Web Development: Created the [CLIMAGS web app](#) to communicate complex climate impact scenarios to stakeholders. **Consulting and Training:** Provided hands-on support and training to researchers on best practices for scientific software development, data visualization, FAIR data principles and reproducible workflows.

2018 – 2023 PhD & Postdoctoral Researcher in Solid State Physics

Quantum matter and spectroscopy

Laboratory for Quantum Matter Research, University of Zurich

Synchrotron Experiments: Angle-resolved photoemission spectroscopy.

Software Development: Created the Python [data-slicer](#) package for handling the ever increasing demands on data analysis and visualization. **High**

Performance Computing: Set up and operated a SLURM-managed HPC environment for density functional theory calculations. **Teaching:** Helped prepare coursework, led exercise classes, supported lecturers.

- 2016 – 2018 Master of Science in Condensed Matter Physics**
University of Zurich with thesis at the University of Amsterdam
Thesis title: Temperature dependent and time resolved photoluminescence of MoTe_2 , MoS_2 and WSe_2
Condensed Matter Physics: Took courses on density functional theory, group theory, superconductors. **High Performance Computing:** Used MPI to parallelize hydrodynamic simulations. **Teaching:** Led exercise classes and supervised lab courses.
- 2015 Junior Scientist at CERN**
Development of a medical PET scanner as member of the [SAFIR](#) project
Data Acquisition and Analysis: Characterized different ASIC designs by comparing performance under standardized measurement setups. **HPC and Modeling:** Simulated PET scanner design using SAGE/Géant. **Software Development:** Employed Python and C++ to create a web interface to a MongoDB database. Co-developed group webpage in Python Flask.
- 2011 – 2014 Bachelor of Science in Physics**
University of Zurich
Thesis title: Measurement of the ζ -Potential in Nanofluidic Devices
 Followed the basic physics curriculum with a focus on **computational approaches and statistics** using Python and Matlab. Minored in **Computer Science**, obtaining a thorough understanding of the mathematical and technological foundations of computers and learning Java.

Skills and Interests

Biogeoscience	Development and application of ecosystem models, meteorological and climatological analysis, grassland productivity
Physics	Angle-resolved photoemission spectroscopy, density functional theory, cuprate high-temperature superconductors, heavy Fermion systems
Computational methods and AI	High performance computing (SLURM), parallelisation (MPI), simulations, machine and deep learning, visualization, data analysis, automation
Programming and DevOps	python, R, matlab, Java, C++, HTML, CSS, Linux/UNIX, bash, Linux package management, $\text{\LaTeX}$, database systems (mongodb), git, CI/CD, Docker
Experimental work	Assembly of a fluorescence microscopy setup, assembly of a cryo-photoluminescence measurement setup, 20+ ARPES beamtimes
Languages	German (native speaker), English (fluent), French (basic), Italian (basic)

Services

- since 2023 Board member [Chorgass-Chor Zürich](#)
- since 2021 Referee for the Journal of Open Source Software
- since 2021 Fire Brigade Obersiggenthal
- 2021 Founding member of the Institute of Physics Sustainability Committee
- 2020 – 2022 Active member of Scientists 4 Future Switzerland
- 2018 – 2021 Junior Researcher representative in the Faculty Assembly at UZH
- Spring 2018 Civil Service: *Served on an alpine pasture. Construction of fences, leading and milking of cows, pasture maintenance.*
- 2010 Compulsory Military Service: *Served as paramedic and truck driver.*

Talks, Conferences and Workshops

1. Contributed talk at The European Geosciences Union's General Assembly 2024: *Integrating Sentinel-2 information into a growth model for assessing Alpine grassland dynamics under climate change*
2. Contributed talk at The International Workshop on Strong Correlations and Angle-Resolved Photoemission Spectroscopy (CORPES) 2019: *ARPES and DFT on overdoped single-layer cuprates*
3. Participation in the Materials with Novel Electronic Properties (MaNEP) Winter School 2019
4. Contributed poster at The 12th International Conference on Materials and Mechanisms of Superconductivity and High Temperature Superconductors (M²S) 2018: *Comprehensive Band Structure Study of Single-layer Cuprate Superconductors*

Publications

1. “growR: R Implementation of the Vegetation Model ModVege”. Kevin P. Kramer and Pierluigi Calanca. 2024. DOI: [10.21105/joss.06260](https://doi.org/10.21105/joss.06260). URL: <https://joss.theoj.org/papers/10.21105/joss.06260>.
2. “Revealing the Orbital Composition of Heavy Fermion Quasiparticles in CeRu₂Si₂”. Kevin P. Kramer, Rina Tazai, Karin von Arx, Masafumi Horio, Julia Küspert, Qisi Wang, et al. 2023. DOI: [10.7566/JPSJ.92.104701](https://doi.org/10.7566/JPSJ.92.104701). URL: <https://journals.jps.jp/doi/10.7566/JPSJ.92.104701>.
3. “Evidence for a Square-Square Vortex Lattice Transition in a High- T_c Cuprate Superconductor”. D. J. Campbell, M. Frachet, S. Benhabib, I. Gilmudinov, C. Proust, T. Kurosawa, et al. 2022. DOI: [10.1103/PhysRevLett.129.067001](https://doi.org/10.1103/PhysRevLett.129.067001). URL: <https://link.aps.org/doi/10.1103/PhysRevLett.129.067001>.

4. “Pseudogap Suppression by Competition with Superconductivity in La-based Cuprates”. J. Küspert, R. Cohn Wagner, C. Lin, K. von Arx, Q. Wang, K. Kramer, et al. 2022. DOI: [10.1103/PhysRevResearch.4.043015](https://doi.org/10.1103/PhysRevResearch.4.043015). URL: <https://link.aps.org/doi/10.1103/PhysRevResearch.4.043015>.
5. “Electronic Reconstruction Forming a C2-symmetric Dirac Semimetal in Ca₃Ru₂O₇ - Npj Quantum Materials”. M. Horio, Q. Wang, V. Granata, K. P. Kramer, Y. Sassa, S. Jöhr, et al. 2021. DOI: [10.1038/s41535-021-00328-3](https://doi.org/10.1038/s41535-021-00328-3). URL: <https://www.nature.com/articles/s41535-021-00328-3>.
6. “Visualization of Multi-Dimensional Data – The Data-Slicer Package”. Kevin Kramer and Johan Chang. 2021. DOI: [10.21105/joss.02969](https://doi.org/10.21105/joss.02969). URL: <https://joss.theoj.org/papers/10.21105/joss.02969>.
7. “Oxide Fermi Liquid Universality Revealed by Electron Spectroscopy”. M. Horio, K. P. Kramer, Q. Wang, A. Zaidan, K. von Arx, D. Sutter, et al. 2020. DOI: [10.1103/PhysRevB.102.245153](https://doi.org/10.1103/PhysRevB.102.245153). URL: <https://link.aps.org/doi/10.1103/PhysRevB.102.245153>.
8. “Band Structure of Overdoped Cuprate Superconductors: Density Functional Theory Matching Experiments”. K. P. Kramer, M. Horio, S. S. Tsirkin, Y. Sassa, K. Hauser, C. E. Matt, et al. 2019. DOI: [10.1103/PhysRevB.99.224509](https://doi.org/10.1103/PhysRevB.99.224509). URL: <https://link.aps.org/doi/10.1103/PhysRevB.99.224509>.
9. “Three-Dimensional Fermi Surface of Overdoped La-Based Cuprates”. M. Horio, K. Hauser, Y. Sassa, Z. Mingazheva, D. Sutter, K. Kramer, et al. 2018. DOI: [10.1103/PhysRevLett.121.077004](https://doi.org/10.1103/PhysRevLett.121.077004). URL: <https://link.aps.org/doi/10.1103/PhysRevLett.121.077004>.
10. “Two-Dimensional Type-II Dirac Fermions in Layered Oxides”. M. Horio, C. E. Matt, K. Kramer, D. Sutter, A. M. Cook, Y. Sassa, et al. 2018. DOI: [10.1038/s41467-018-05715-2](https://doi.org/10.1038/s41467-018-05715-2). URL: <https://www.nature.com/articles/s41467-018-05715-2>.
11. Robert Becker, Jean-Pierre Cachemiche, Chiara Casella, Günther Dissertori, Jannis Fischer, Alexander S. Howard, et al. “Monte-Carlo Simulation Based Estimation of NECR, Sensitivity, and Spatial Resolution of a Novel Preclinical PET Insert for MR”. In: *2015 IEEE Nuclear Science Symposium and Medical Imaging Conference (NSS/MIC)*. Oct. 2015, pp. 1–3. DOI: [10.1109/NSSMIC.2015.7582128](https://doi.org/10.1109/NSSMIC.2015.7582128). URL: <https://ieeexplore.ieee.org/document/7582128> (visited on 05/06/2024).